

Glossary of General Relativity Terminology

Supplementary Material for: “Twins in Relativistic Spacetimes: Dispelling Some Misconceptions”

T. A. A. da Câmara

Instituto de Física, Universidade Federal do Rio de Janeiro, Brazil

P. S. Lemos

Instituto de Física, Universidade Federal do Rio de Janeiro, Brazil

M. O. Calvao

Instituto de Física, Universidade Federal do Rio de Janeiro, Brazil

F. D. Sasse

Centro de Ciências Tecnológicas, Universidade do Estado de Santa Catarina, Brazil

This glossary provides some definitions of key concepts in general relativity and differential geometry used in the accompanying article.

- *manifold*: A set of points (events) locally equivalent to some Euclidean N -dimensional space [2, p. 11].
- *chart*: A bicontinuous bijection from a subset of a manifold to an open subset of $\mathbb{R}^N$ [2, p. 12], [3, p. 4].
- *atlas*: A collection of charts covering the manifold [3, p. 19].
- *congruence*: A family of curves, defined on a subset $\mathcal{U}$ of the manifold, such that each event in $\mathcal{U}$ belongs to one and only one curve [2, p. 216].
- *coordinate system*: A chart [2, p. 12], [3, p. 3].
- *spacetime*: A pair $(\mathcal{M}, g_{\alpha\beta})$, where $\mathcal{M}$ is a differentiable manifold and $g_{\alpha\beta}$ is a Lorentzian metric tensor field defined on $\mathcal{M}$ [1, p. 17].
- *observer*: A timelike future-directed normalized curve [1, p. 36].
- *instantaneous observer*: A pair $(\mathcal{P}, u_{\mathcal{P}}^{\alpha})$, where $\mathcal{P}$ is an event of the manifold $\mathcal{M}$ and $u_{\mathcal{P}}^{\alpha}$ is a timelike future-directed normalized tangent vector (four-velocity) in $T_{\mathcal{P}}\mathcal{M}$ [1, p. 41].
- *reference frame*: A vector field whose integral curves are observers [1, p. 52].
- *Killing vector field*: An infinitesimal symmetry of the metric tensor (isometry), i.e., a solution $K^{\alpha}(x^{\beta})$ of Killing's equation: $K_{(\alpha;\beta)} = 0$ [2, p. 441], [1, p. 81].
- *tetrad*: An (orthonormal) basis $e_{(A)}^{\alpha}$ for the tangent space at a given event [2, p. 49].
- *static spacetime*: A spacetime with a timelike Killing vector field that is also hypersurface-orthogonal [2, p. 119].

References

- [1] R. K. Sachs and H. H. Wu, *General Relativity for Mathematicians*, Springer, New York (1977), 181 pp.
- [2] R. M. Wald, *General Relativity*, University of Chicago Press, Chicago (1984), 491 pp.
- [3] J. M. Lee, *Introduction to Smooth Manifolds*, 2nd edition, Springer, New York (2013), 708 pp.